01 Getting Started: How, Why, What

Starting off this semester, in this module, my main goal is to expand my creative practice.

How
How would I like to tackle this work? With a mix of co-design, communication, writing, and AI. I'm excited about the new tools that are being created every day and their capacity to expand what's possible for me to create.

Why
What drives me? I really enjoy the process of making functional art. Any type of functional design excites me: ceramics, sewing, architecture, UX design. But almost all of that work has been for me or academic projects where the design never actually got used by anyone but me and my marker. The closest I've come to designing for real users is when I taught English to elementary school kids. I was designing lessons for a target user, crafting things that were fun and beautiful for a very specific purpose. That work is what I felt most fulfilled by, and I want more of it.

What
What do I want to learn? How to take a vague brief and translate it into something interesting and creative. I'm particularly looking forward to working with this The Natural History Museum brief because it's a real audience, a real institution, and a real problem.

A Note from the Future
I wrote this blog post right at the start of the semester. Now, as I'm sitting down to compile the blog, I've decided to put some of what I've learned to the test. Not only will this blog be a practice in reflecting and communicating my design work, I also have decided to vibe code this blog site as a way to continue working with and learning how to incorporate AI into my work. Stay tuned for more reflection on my experience using AI throughout the Living Record Project and in the creation of this blog.

02 Understanding the Problem Space

Before we could begin designing anything, my team and I needed to understand the problem space. We started by individually compiling a set of sources about the Natural History Museum gardens, the Urban Nature Project, and the museum's broader work, and uploaded them into NotebookLM. From there, we each generated a podcast: a quick, conversational summary of the material we had read. This was a useful way to surface the key points.

Once we had a shared baseline, we ran a more structured exercise. Each group member wrote down 20 answers to the question: *What is the function of the Natural History Museum gardens?* Working alone first was important, because it kept us from converging too early on the obvious answers. Together, that produced 60 statements ranging from "an education center" to "a wildlife research site."

We then carried out a thematic analysis of the full set of 60, grouping similar answers, identifying patterns, and naming the resulting clusters. Five overarching themes came out of the analysis: **Education, Leisure, Research, Museum Mission, and Environmental**. Each one captured a different aspect of the garden's relationship between the museum, the visitors, and the space itself.

To explore the themes further, we built a moodboard for each. I had expected the moodboards to be a one off output of the discovery phase, but throughout the rest of the project, we kept returning to specific images from those boards, using them as references for individual rooms, interactions, and design decisions. The themes told us what the gardens were about and the moodboards gave us a visual library to draw inspiration from.

03 Defining the problem space

(hero image: define double diamond)

When I last left off, we had 60 answers to the question "What is the function of the NHM gardens?" and five overarching themes to show for it. We had a broad, shared understanding of the problem space. But understanding something broadly and knowing what to do about it are very different things.

To begin generating ideas, we tested out different ideation techniques. Sydney ran the critical thinking with AI exercise from class, and the outputs she shared caused us to engage with perspectives we hadn't considered: missing stakeholders, underlying assumptions, and questions we hadn't thought to ask.

For further ideation, Tom encouraged us to try techniques beyond what we'd done in other modules. We decided on a brainwriting exercise. Unlike Crazy 8s, which is individual, brainwriting is collaborative and sequential. Each person started writing ideas on a piece of paper, then passed it to the next person after two minutes where you pick up where someone else left off, building on or departing from their ideas. We did two full cycles (twelve minutes total), reviewed what we had, then ran a third focused cycle of six minutes to hone in on the stronger threads. (image: brainwrite2 with caption)

I found it difficult to push myself to generate new ideas in such short bursts of time. But the combination of my teammates' ideas already on the page and the time limit worked together to force divergent thinking. There was no time to stall or wait for ideas to come.

After all three cycles, we put our ideas on sticky notes and clustered them by theme. (image: brainwrite affinity diagram with caption) Categories emerged: digital twins, interactive experiences, data visualisation, VR. Sitting with those clusters, the ideas that excited us most were the ones that put visitors inside the data. An exhibit. A physical, immersive, interactive exhibit inside the Natural History Museum itself.

More on what that meant for our design direction in the next post.

04 TITLE

(hero image: problem statement double diamond)

(image: lightblub icon) Problem statement: *How might we create an exhibit that allows visitors to interact with the collected ecological data, the gardens, and with the museum itself?*

Before we could design anything, we needed to understand what good exhibit design looked like in practice. We started with research. The Smithsonian's exhibit development framework gave us a useful structure. It centers on a single big idea, supported by key messages and critical questions that the exhibit should answer for visitors.

(image: ipop model with caption) Alongside that, we came across the IPOP model, which argues that visitors come to museums with four different orientations: some are drawn to Ideas (abstract thinking), some to People (emotional connection), some to Objects (visual aesthetics), and some to Physical experiences (multisensory engagement). A well-designed exhibit has to work for all four types at once. That framing became a lens we kept returning to throughout the rest of the project.

But research only gets you so far. We needed to go out and see things.

(image: Outernet movie and outernet inspo images with caption) The first trip was to Outernet London. It showed us what immersion could actually feel like when it was done well, and gave us an idea of what we wanted to aim for.

(image: dinosaur movie with caption) The second trip was back to the Natural History Museum itself, this time with a structured observation framework. Using the Robson framework as a guide, we catalogued interaction types (touch screen kiosks, physical controls, video screens, static text, animatronics, immersive rooms, physical 3D models, and audio recordings), visitor flow (mix of open exploration spaces and single-route exhibits), and demographics (large school groups, international couples and families) which helped confirm our target audience.

More on how these observations fed into the specifics of our design in the next post.

05 specifying our exhibit part 1: Sketching as a creative practice

(image: develop double diamond)

We had a problem statement, a framework, and a collection of field observations. What we didn't have yet was *the idea*: the organizing logic that would give our exhibit a shape and a story. How we got that was, in part, through sketching.

Sketching as a creative practice has been one of the biggest takeaways for me from this module. I have some prior experience with fine arts (Studio Art minor in undergrad), but I hadn't picked up my sketchbook in years. Richard Banks' lecture on his design practice and the role of sketching inspired me to pick it up again. I wanted to challenge myself to not aim for something polished, but just work on externalizing ideas and thinking with my hands.

So on the bus home after one of our team meetings, I took out my notebook and just started drawing some very messy, barely comprehensible sketches (image: micro to macro sketch1 with caption) and somewhere in those scrawls, an idea surfaced: what if the exhibit started at the smallest possible scale (DNA, microorganisms) and then gradually zoomed out, room by room, to the garden, then to London, then to the global picture? *Micro to macro.*

The next time our team met, we sat down to formalize our exhibit concept using the Smithsonian framework as a guide. We locked in the type of exhibit (permanent and research led), our target audience (international visitors), our purpose statement (to propose a novel way for visitors to interact with the NHM's ecological data), and our main message (changes in these gardens are representative of larger changes in the world). (have this section as a table with type of exhibit, purpose statement, main message on the left and the corresponding decisions on the right)

When it came to narrative organization, I shared the micro to macro idea from my sketch, and it fit. International visitors aren't necessarily interested in one garden in South Kensington, but they are interested in what that garden says about the planet they live on. The structure gave our exhibit a reason to exist beyond the museum's walls.

Sketching will be a tool that myself and my teammates continue to use throughout the process. More on that in the next post.

06 Specifying our exhibit part 2: getting into the details

(image: develop double diamond)

Once we had the broad strokes, (a permanent, research led exhibit with a micro-to-macro narrative aimed at international visitors) it was time to get into the specifics. What would actually be *in* this exhibit? What would visitors touch, see, and do?

We used a sequence of ideation and specification tools to work this out, each one building on the last.

First, a round of Crazy 8s with the aim of generating specific interactions, content types, and moments within rooms. From there, we used the ideas from Crazy 8s to create sketchnotes. Sketchnotes pushed me to slow down and think through a single idea more fully by combining text, image, and annotation in a way that surfaces more details than the Crazy 8s sketches. My sketchnotes helped me work out the logic of the exhibit’s interactive elements like an acoustic data visualization, or touch screen interaction with garden habitat elements. (image: sketch-note - have this image inline with the text and with a caption and have it clickable, on hover have it move a little to show clickability, and on click have it open an overlay to show larger and in the overlay allow for zoom and pan around)

We then created storyboards(image: story-board with caption) and a user journey map to understand how all these interactions would connect. Mapping the journey from a visitor walking into the exhibit to walking back out made the gaps visible. What specifically were the different rooms and spaces within the exhibit? What interactions would inhabit each one?

Those gaps fed directly into our specification session, where we sat down as a team and defined every room in the exhibit, aligned with our micro-to-macro narrative. The rooms ran from context about the gardens and the Urban Nature Project, down to the DNA and microbe level, up through the garden itself, then out to London and the global picture, ending with a call to action to visit the gardens.(image: exhibit flow with caption)

At this time, we also decided to focus the majority of our design efforts around the central garden room (garden room sketch with caption). Through looking back at our moodboards, team meeting notes, and lengthy discussion, we came up with the three core interactive components for the garden room: a QR code language selector, a weather data control panel, and a set of habitat kiosks pairing physical models with digital exploration tools. (image: nature kiosk sketch, weather data sketch with caption)

Next up: building it.

07 Building our exhibit

(hero image: deliver double diamond)

With a fully specified exhibit and a clear focus on the garden room, it was time to build.

(full width movie of of the living record exhibit movie, with autoplay and player controls)

Three principles guided everything we built: translation access (the exhibit needed to work for international visitors who might not read English), visual storytelling (data should be felt before it's explained), and intuitive navigation. (images: translation-services-sketch, image-focused-sketch, intuitive-navigation-sketch)

We divided the work by skill and interest. Sydney built the 3D model of the entire exhibit in Blender. To render the walls of each room, we worked as a team using Adobe Firefly to generate images that reflected the data visualisations we'd specified: a falling-species eDNA visualisation for the microbe room, an acoustic sound-wave artwork for the audio level room. (images: wall1, wall2, wall3 small images)

Sydney then made the model clickable and interactive. Yemisi and I built the garden room prototypes: Yemisi handled the language selector and the date picker for the weather control panel, and I built the habitat kiosk. I then used Gemini and Figma to place both interactions into the garden room landscape. (images: control panel movie with caption, kiosk prototype demo with caption)

This stage pushed me to develop real strategies for AI image generation, thinking carefully about dimensions, output style, and data accuracy. Getting an AI to produce something that reads as a genuine data visualisation rather than a decorative mush requires very deliberate prompting. Speaking of AI as a creative tool: a quick meta note on this blog itself. Since our vibe coding session with Richard, I've been looking for more ways to practice this technique.

I decided to vibe code this blog site rather than using Wix or a similar no-code platform, and that decision has given me a level of creative control I wouldn't have had otherwise, like in the hero section on the main page, where I was able to add in micro-interactions to illustrate different aspects of the project. It has felt like an extension of what we've discussed about AI throughout this course, using it not as a shortcut, but as a way to be creative beyond my own technical limitations.

Next up: the final presentation and what I learned from communicating our work.

08 Final: Presentation and Takeaways

(hero image: deliver double diamond)

At the start of this module, I wrote about what I was hoping to get from it: a more intentional design practice, experience working in a real-world brief context, and a better understanding of how to use AI as a creative tool. Eleven weeks later, with a finished exhibit proposal and a final presentation delivered at the Natural History Museum itself, I want to reflect on whether that happened as well as other unexpected takeaways.

Starting with the final presentation, which was a design problem in and of itself. Tom's lecture on communication in design framed it well: the medium is the message. The way you choose to communicate your work shapes how it's received, regardless of the quality of the work itself. That meant thinking carefully about who our audience was (the NHM research and UX team), what we needed them to understand (the design and the process behind it), and what medium would carry that most effectively. (image: presentation)

Despite low attendance from the NHM team, presenting there felt meaningful. The feedback that our presentation and deliverables were of professional quality was genuinely gratifying not just of the work that we'd done, but that we were communicating what we intended it to communicate.

What I've found through writing this blog is a unifying thread: adding structure to freedom as a means to promote creativity. The tools and frameworks we used: the double diamond, the Smithsonian exhibit structure, the IPOP model, the Robson framework, brainwriting, were all scaffolding that made creativity possible. Without them, a brief this open would have stayed open. The skills I'm taking away feel the same: sketching as a thinking tool, deliberate AI prompting, vibe coding, communicating design, all of them are ways of providing structure and support in the right places and expanding what I'm able to make because of it.

(image: NHM-presenation)

I came into this module wanting to be a more intentional designer. I think I'm leaving as one.